

SECTION B

7 (short)

- (a) Explain the difference between the stress concentration factor and the stress intensity factor. Define the critical stress intensity factor. [4]
- (b) A thin-walled, long cylindrical vessel of 2 m diameter is to hold an internal pressure of 10 MPa. Non-destructive testing revealed the presence of small microscopic cracks in the inner wall. The cracks are expected to gradually extend through the pressurised wall by fatigue. The fracture toughness of the vessel material is $110 \text{ MPa}\sqrt{\text{m}}$. Find the minimum wall thickness of the vessel that can be used while ensuring that the vessel will not fail by fast fracture. Assume the dimensionless constant $Y = 1$. [6]